

Migraciones:Relación de uno a muchos en Django con MySQL

Aprendiz: *Sebastian Sanchez Martinez*

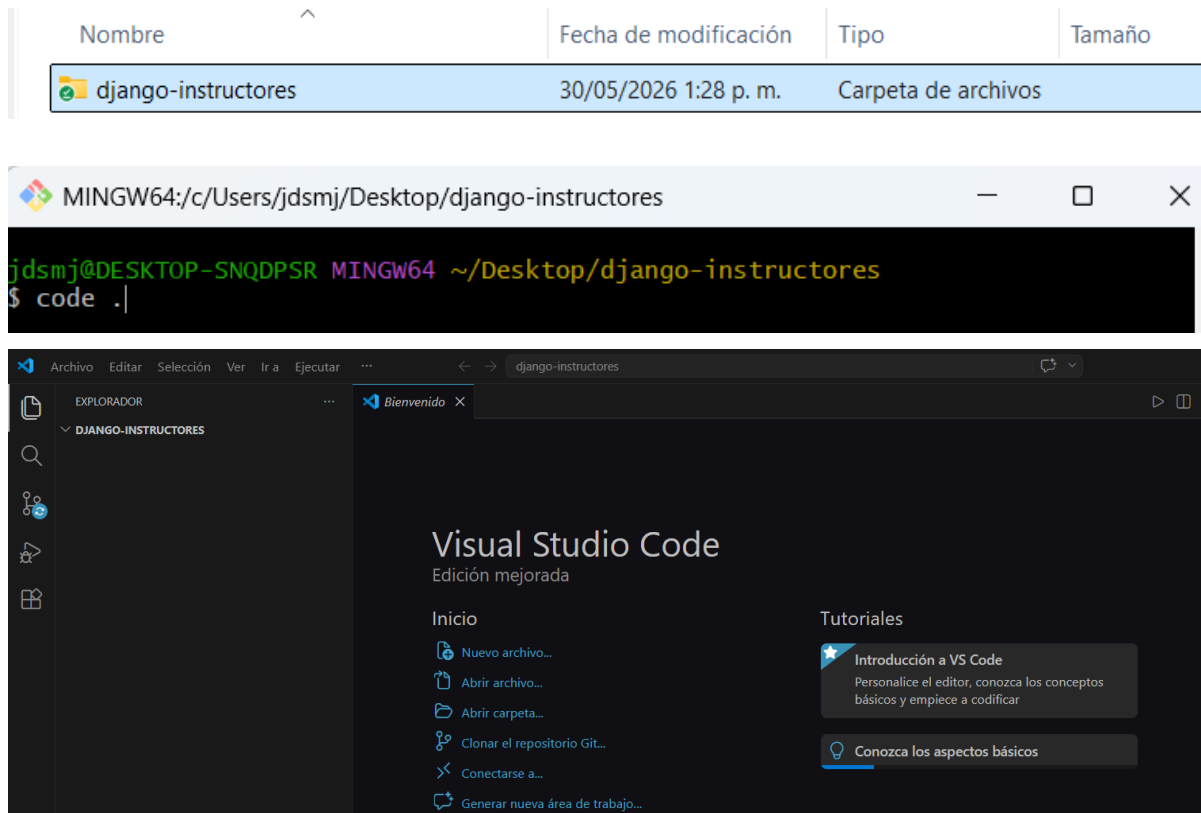
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Fecha: 30/05/2026

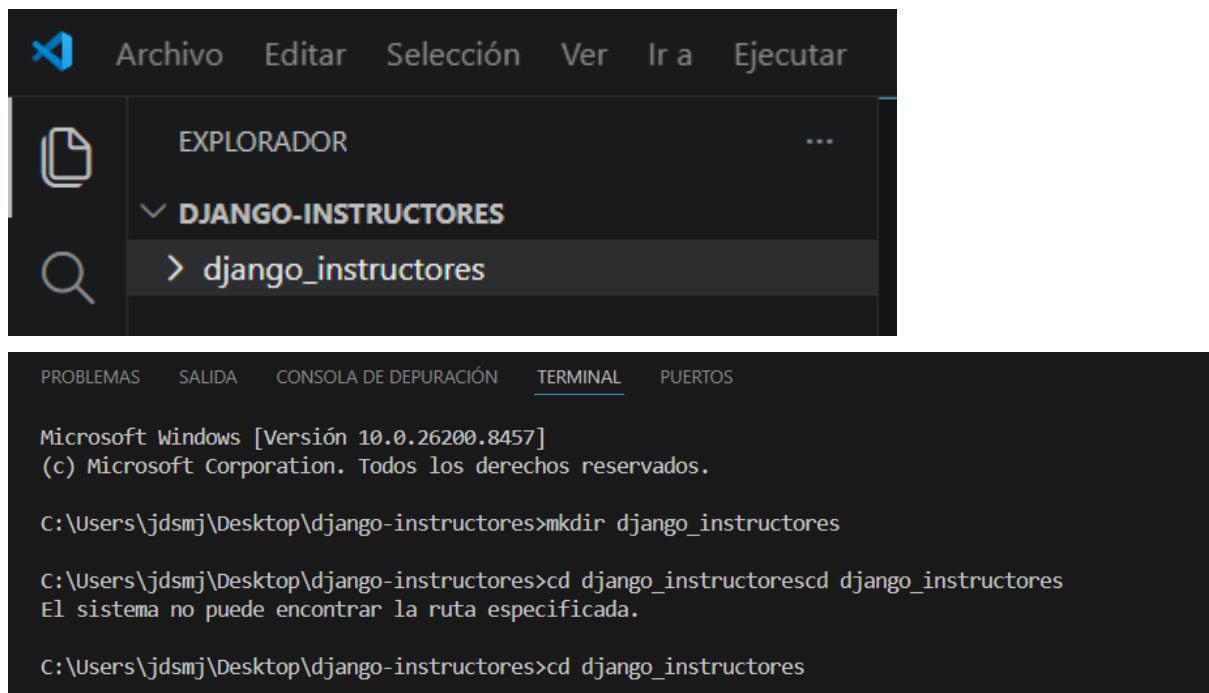
Instructor: *Alexa Muñoz*

Programa: *Analisis y desarrollo de software*

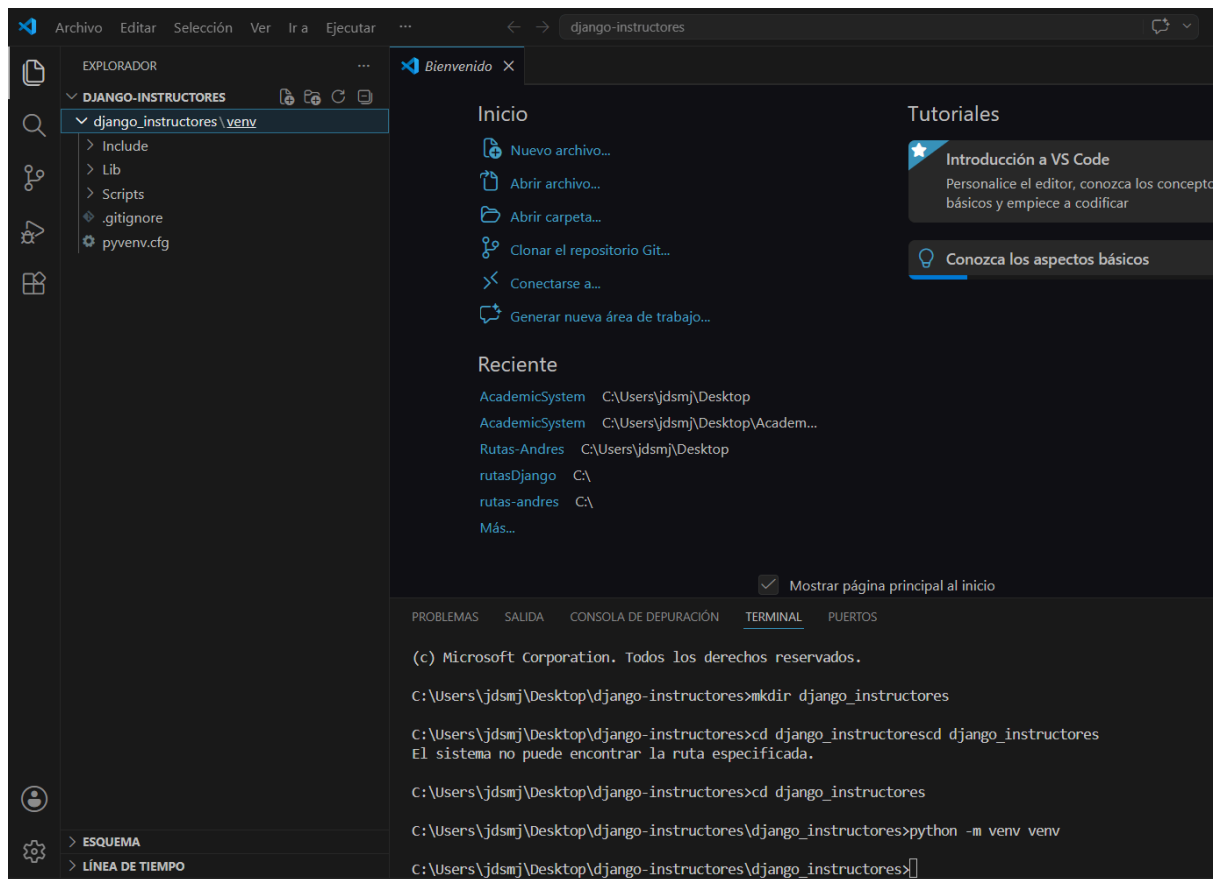
1. CARPETA CREADA EN VS CODE:



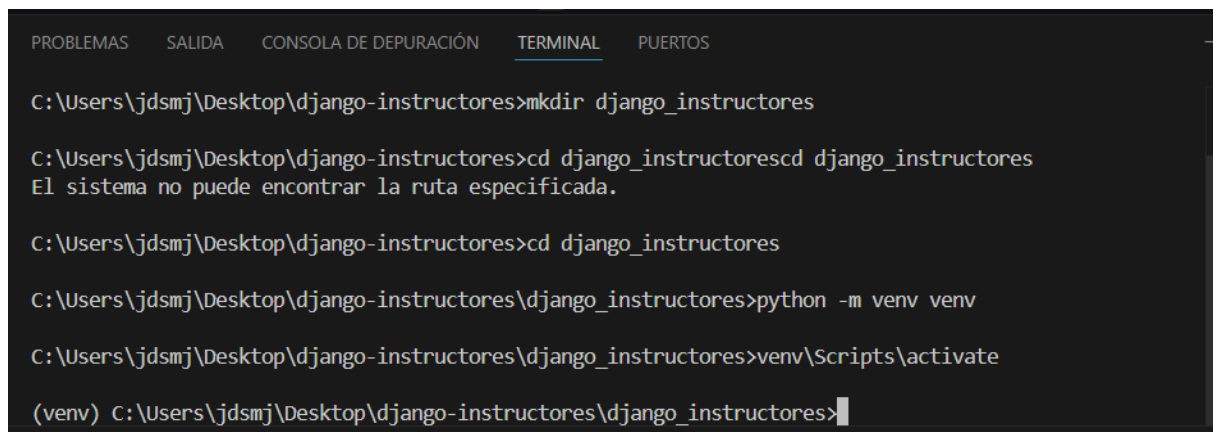
2. CREAR LA CARPETA DEL PROYECTO:



3. CREAR EL ENTORNO VIRTUAL:



4.ACTIVAMOS EL ENTORNO:



5.INSTALAMOS DJANGO:

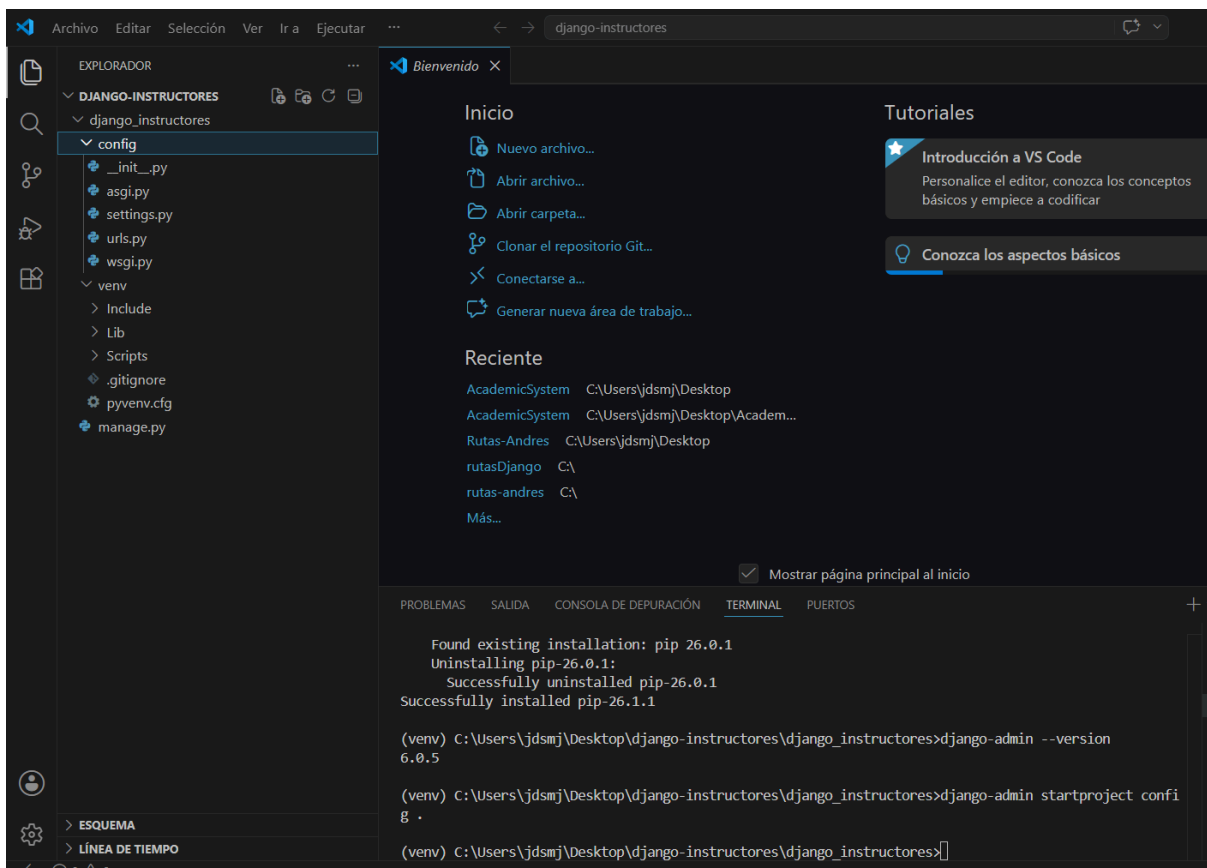
```
PROBLEMAS  SALIDA  CONSOLA DE DEPURACIÓN  TERMINAL  PUERTOS

(venv) C:\Users\jdsjmj\Desktop\django-instructores\django_instructores> python.exe -m pip install --up
grade pip
Requirement already satisfied: pip in .\venv\Lib\site-packages (26.0.1)
Collecting pip
  Using cached pip-26.1.1-py3-none-any.whl.metadata (4.6 kB)
Using cached pip-26.1.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
  Attempting uninstall: pip
    Found existing installation: pip 26.0.1
    Uninstalling pip-26.0.1:
      Successfully uninstalled pip-26.0.1
```

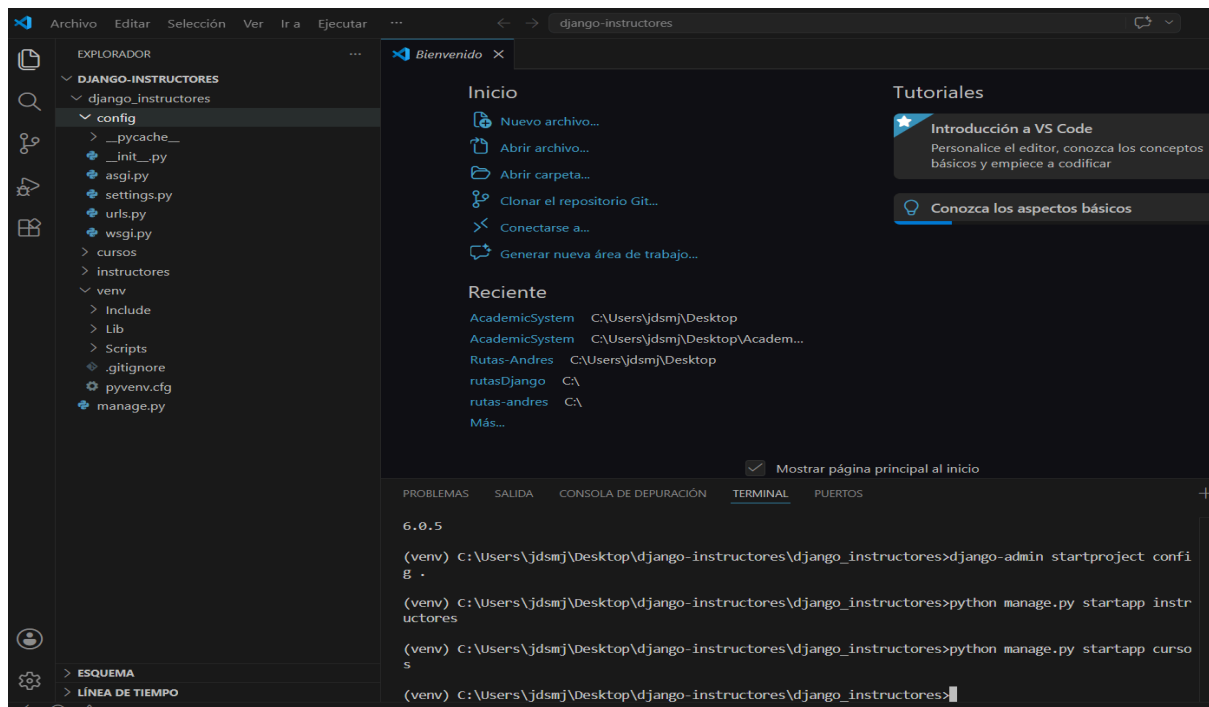
6. VERIFICAMOS:

```
(venv) C:\Users\jdsjmj\Desktop\django-instructores\django_instructores>django-admin --version
6.0.5
```

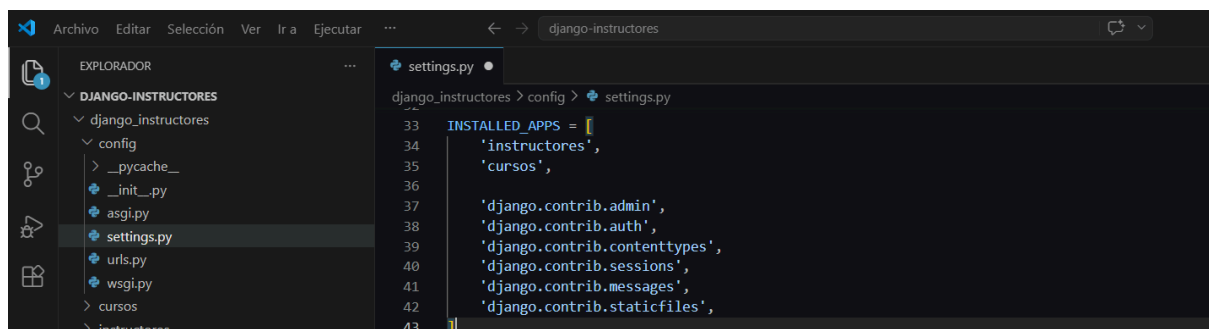
7. CREAMOS EL PROYECTO:



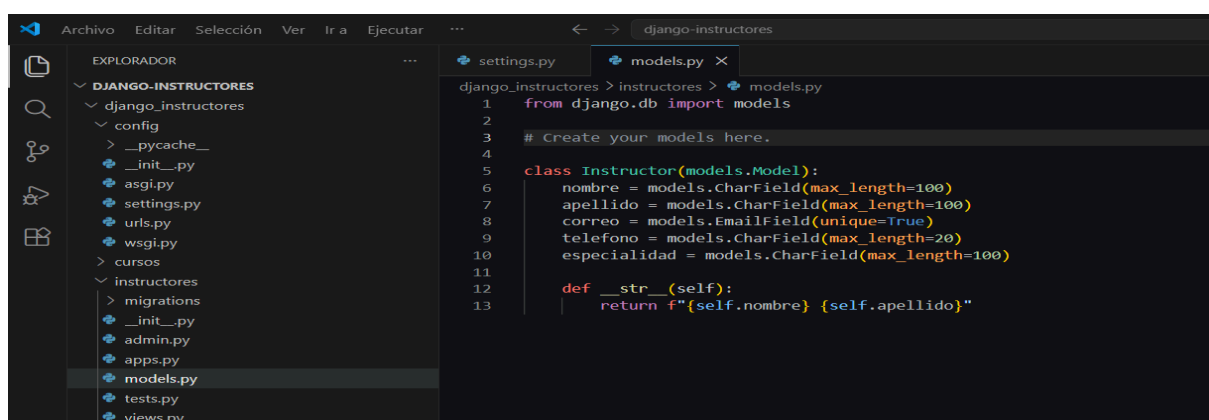
8.CREAMOS LAS APLICACIONES QUE NECESITA NUESTRO PROYECTO:



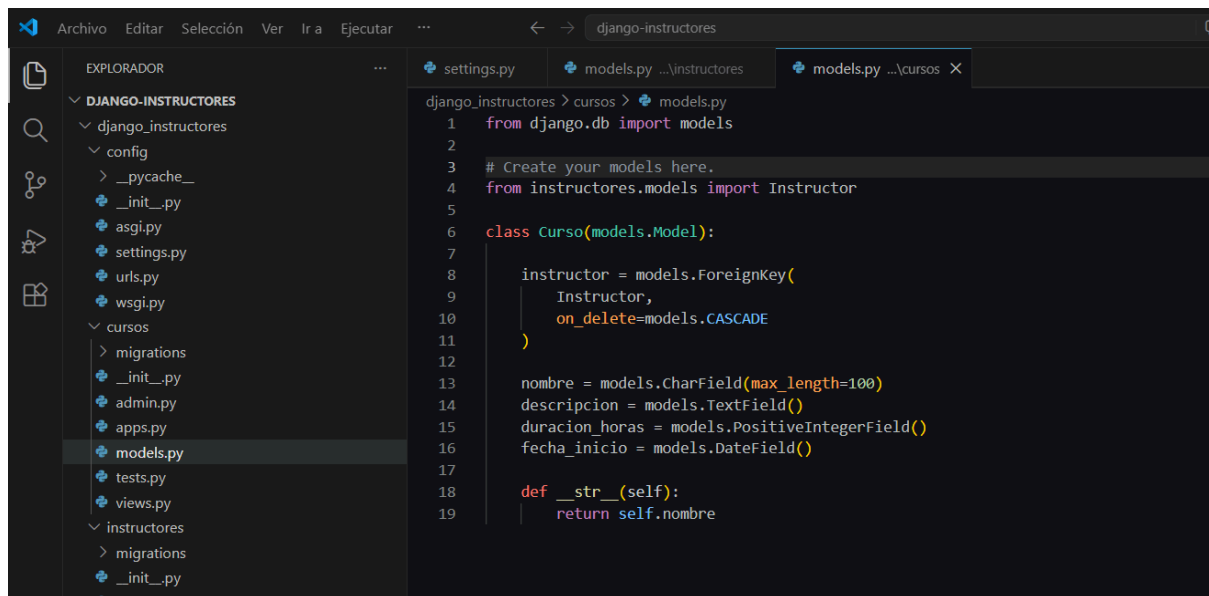
9.VAMOS A CONFIG/SETTINGS.PY:



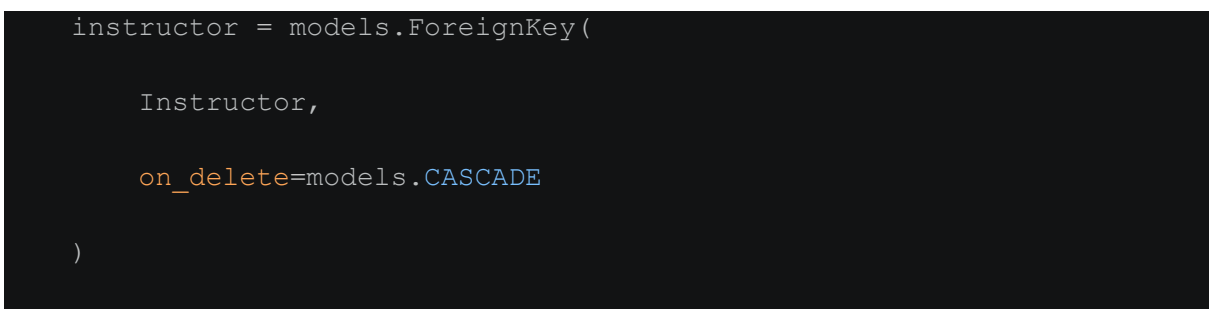
10.LUEGO INSTRUCTORES/MODELS.PY:



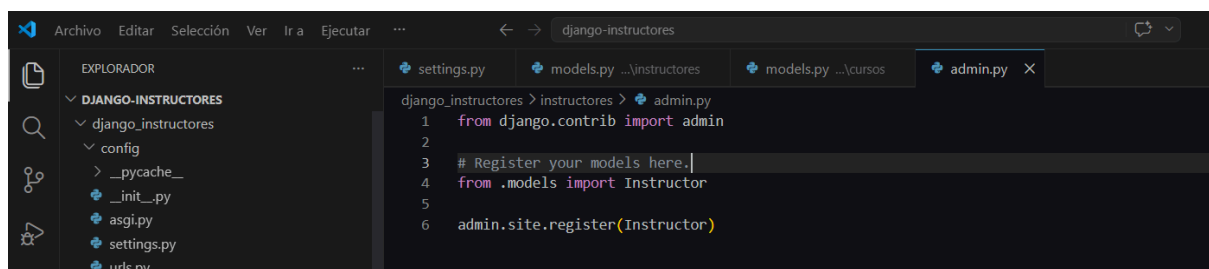
11.LUEGO A CURSOS/MODELS.PY:



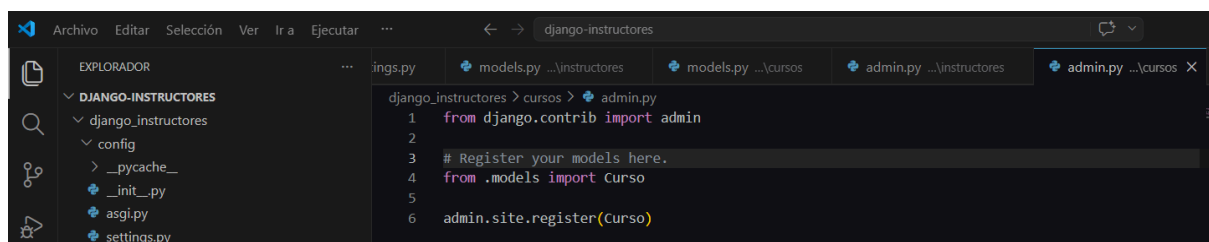
Y AQUI MUESTRA LA PRINCIPAL RELACIÓN DE INSTRUCTOR 1-> MUCHO RECURSOS:



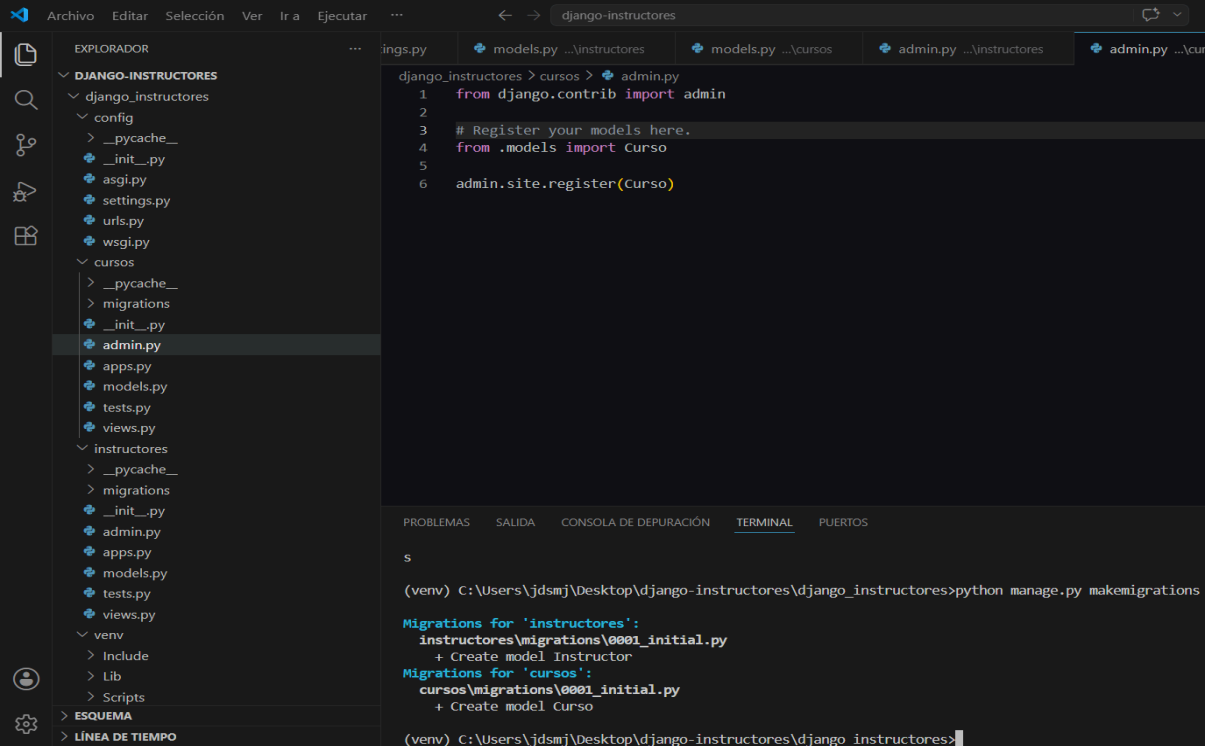
12.REGISTRAMOS LOS MODELOS EN ADMIN INSTRUCTOR/ADMIN:



13.REGISTRAMOS LOS MODELOS EN ADMIN CURSO/ADMIN:



14.CREAMOS LAS MIGRACIONES:



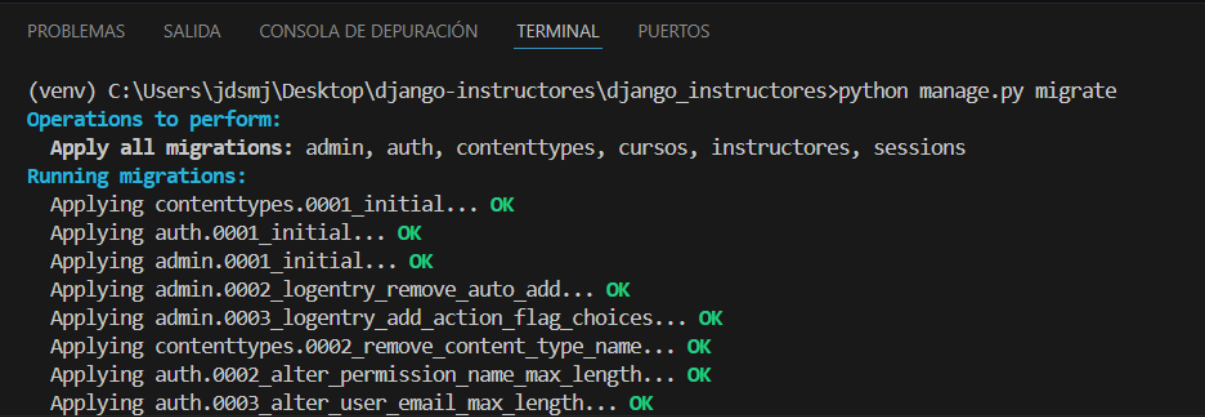
The screenshot shows the Visual Studio Code interface with a Django project named 'django-instructores'. The Explorer sidebar on the left shows the project structure, including 'django_instructores', 'config', 'cursos', and 'instructores'. The 'admin.py' file in the 'cursos' directory is selected and open in the editor. The code in 'admin.py' is as follows:

```
1 from django.contrib import admin
2
3 # Register your models here.
4 from .models import Curso
5
6 admin.site.register(Curso)
```

The Terminal at the bottom shows the command 'python manage.py makemigrations' being executed. The output indicates that migrations were created for both the 'instructores' and 'cursos' apps:

```
(venv) C:\Users\jdsmj\Desktop\django-instructores\django_instructores>python manage.py makemigrations
Migrations for 'instructores':
  instructores\migrations\0001_initial.py
  + Create model Instructor
Migrations for 'cursos':
  cursos\migrations\0001_initial.py
  + Create model Curso
(venv) C:\Users\jdsmj\Desktop\django-instructores\django_instructores>
```

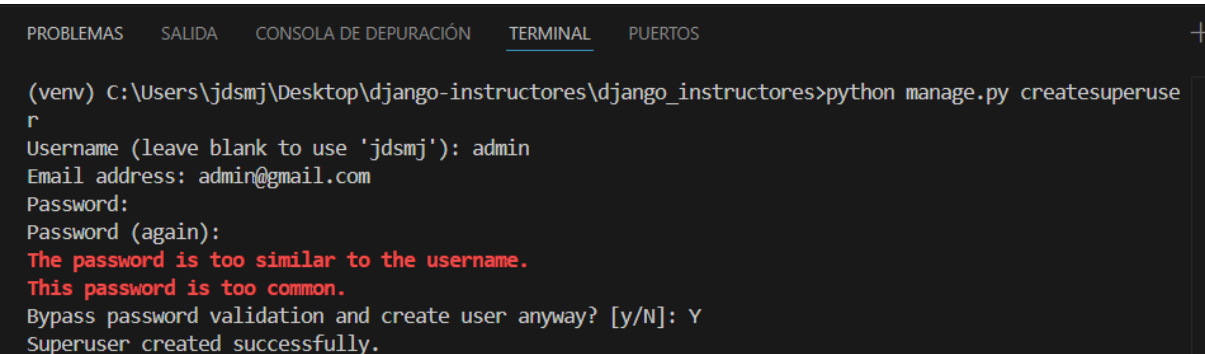
15.EJECUTAR EN MIGRACIONES:



The screenshot shows the terminal output of the command 'python manage.py migrate'. The output indicates that all migrations were applied successfully:

```
(venv) C:\Users\jdsmj\Desktop\django-instructores\django_instructores>python manage.py migrate
Operations to perform:
  Apply all migrations: admin, auth, contenttypes, cursos, instructores, sessions
Running migrations:
  Applying contenttypes.0001_initial... OK
  Applying auth.0001_initial... OK
  Applying admin.0001_initial... OK
  Applying admin.0002_logentry_remove_auto_add... OK
  Applying admin.0003_logentry_add_action_flag_choices... OK
  Applying contenttypes.0002_remove_content_type_name... OK
  Applying auth.0002_alter_permission_name_max_length... OK
  Applying auth.0003_alter_user_email_max_length... OK
```

16.CREAR EL USUARIO:



The screenshot shows the terminal output of the command 'python manage.py createsuperuser'. The output indicates that the superuser was created successfully, although the password was too similar to the username and too common.

```
(venv) C:\Users\jdsmj\Desktop\django-instructores\django_instructores>python manage.py createsuperuser
Username (leave blank to use 'jdsmj'): admin
Email address: admin@gmail.com
Password:
Password (again):
The password is too similar to the username.
This password is too common.
Bypass password validation and create user anyway? [y/N]: Y
Superuser created successfully.
```

17.EJECUTAR EN EL SERVIDOR:

```
PROBLEMAS  SALIDA  CONSOLA DE DEPURACIÓN  TERMINAL  PUERTOS

(venv) C:\Users\jdsjm\Desktop\django-instructores\django_instructores>python manage.py runserver
Watching for file changes with StatReloader
Performing system checks...

System check identified no issues (0 silenced).
May 30, 2026 - 14:40:54
Django version 6.0.5, using settings 'config.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.

WARNING: This is a development server. Do not use it in a production setting. Use a production WSGI o
r ASGI server instead.
```

18.VAMOS A NUESTRO PANEL ADMINISTRATIVO (INSTRUCTOR):

← ↻ ⓘ 127.0.0.1:8000/admin/

Django administration

Site administration

AUTHENTICATION AND AUTHORIZATION	
Groups	+ Add Change
Users	+ Add Change

CURSOS	
Cursos	+ Add Change

INSTRUCTORES	
Instructors	+ Add Change

Recent actions

My actions

None available

Django administration

Home > Instructores > Instructors > Add instructor

Start typing to filter...

AUTHENTICATION AND AUTHORIZATION	
Groups	+ Add
Users	+ Add

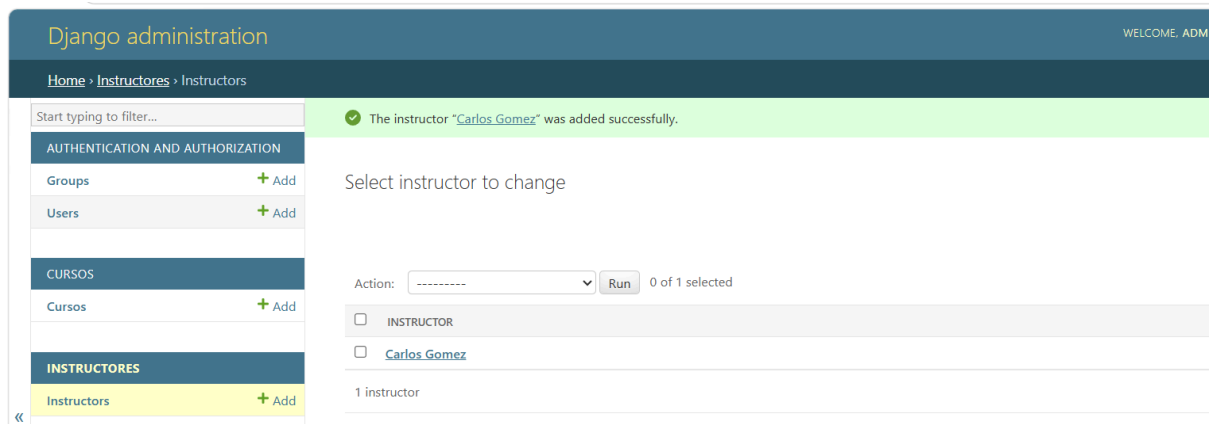
CURSOS	
Cursos	+ Add

INSTRUCTORES	
Instructors	+ Add

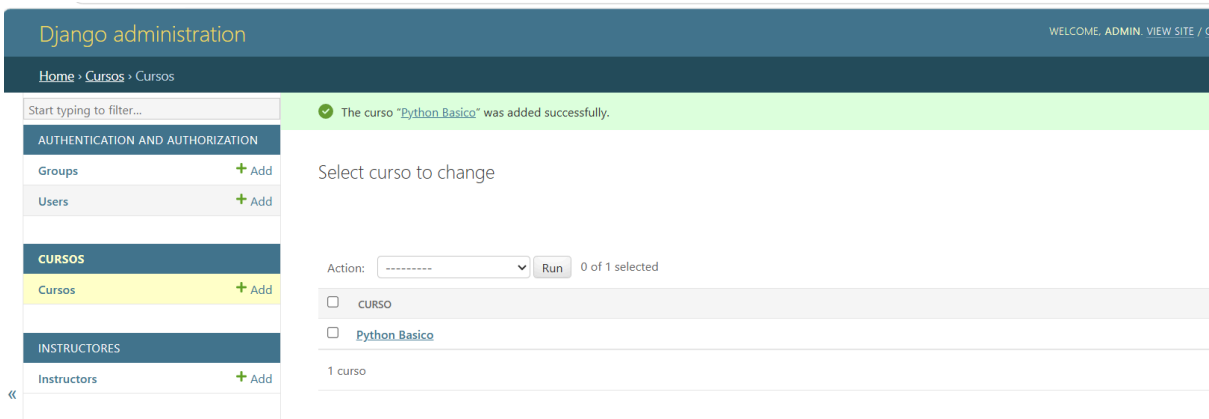
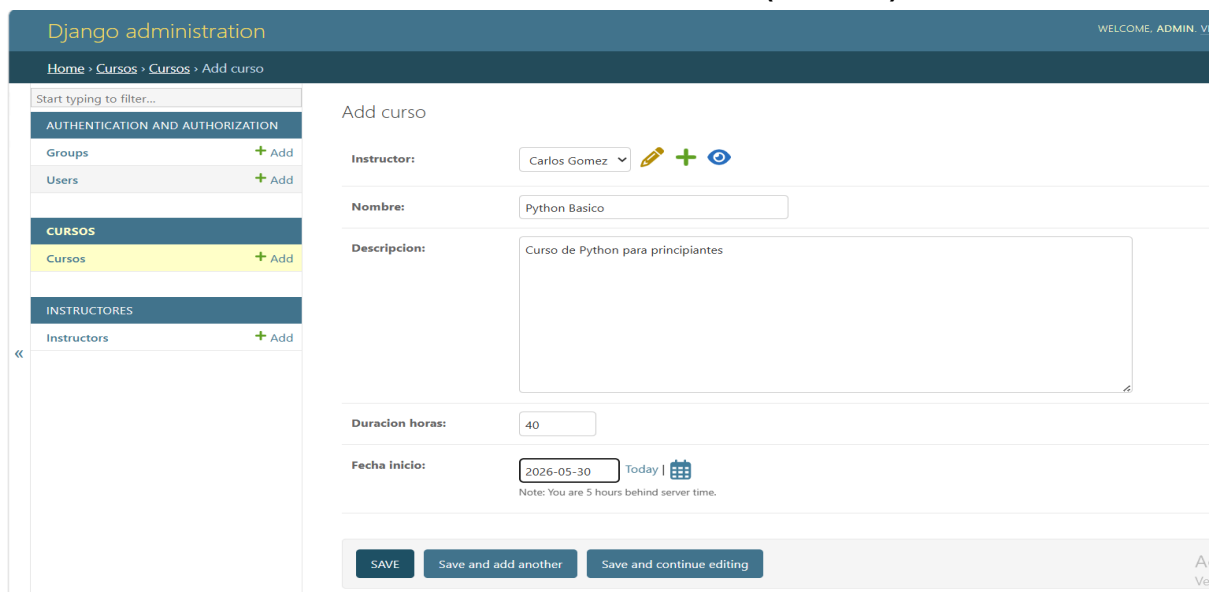
Add instructor

Nombre:	Carlos
Apellido:	Gomez
Correo:	carlos@gmail.com
Telefono:	3001234567
Especialidad:	Desarrollo Web

[SAVE](#) [Save and add another](#) [Save and continue editing](#)



19.VAMOS A NUESTRO PANEL ADMINISTRATIVO (CURSO):



20.CONCLUSIÓN:

En esta actividad se desarrolló un proyecto en Django para implementar una relación de uno a muchos entre las tablas Instructor y Curso. Se crearon los modelos correspondientes, se configuró la relación mediante

una clave foránea (ForeignKey) y se realizaron las migraciones necesarias para generar las tablas en la base de datos.

Además, se verificó el funcionamiento de la relación a través del panel de administración de Django, comprobando que un instructor puede tener varios cursos asociados, mientras que cada curso pertenece a un único instructor. Esta práctica permitió comprender la importancia de las relaciones entre tablas y el uso de ForeignKey para modelar correctamente la información dentro de una aplicación web desarrollada con Django.